Operating Systems - CS(D)

Quiz-7 Solution

Demand-Paged Virtual Memory System

Given:

- 24-bit addressing, 8 KB page size, 8 MB main memory.
- Main memory access time: 100 ns.
- Page fault service time: 10 ms = 10,000,000 ns.
- Page table in main memory.
- TLB access time: 10 ns, TLB hit rate: 0.9 (for parts 2 and 3).

1. Time to Load a Byte (No TLB, Page in Main Memory) (3 marks)

Without a TLB, address translation requires accessing the page table in main memory, followed by accessing the data:

- Page table access: 100 ns.
- Data access: 100 ns.
- Total time: $100 + 100 = 200$ ns.

  Marking Scheme:

  - Correct answer (200 ns): 3 marks.
  - Incorrect or no attempt: 0 marks.

2. Effective Memory Access Time with TLB (4 marks)

With a TLB, calculate effective memory access time (EMAT) considering TLB hit (0.9) and miss (0.1):

- **TLB Hit** (0.9):
  - TLB access: 10 ns.
  - Data access: 100 ns.
  - Total: $10 + 100 = 110$ ns.
- **TLB Miss** (0.1):
  - TLB access: 10 ns.
  - Page table access: 100 ns.
  - Data access: 100 ns.
  - Total: $10 + 100 + 100 = 210$ ns.
- **EMAT**: $(0.9 \times 110) + (0.1 \times 210) = 99 + 21 = 120$ ns.

  Marking Scheme:

  - Correct TLB hit time (110 ns): 1.5 marks.
  - Correct TLB miss time (210 ns): 1.5 marks.
  - Correct EMAT (120 ns): 1 mark.
  - Incorrect or no attempt for any part: 0 marks for that component.

3. Time to Load Page with TLB (Page Fault) (3 marks)

With a TLB and a page fault, the sequence is:

- **Initial TLB Miss**:
  - TLB access: 10 ns.
  - Page table access (detects page fault): 100 ns.
  - Total: $10 + 100 = 110$ ns.
- **Page Fault Service**: $10\,\text{ms} = 10,000,000$ ns.
  Note: The page table and TLB are updated during the page fault service time
- **Retry (TLB Hit)**: TLB updated post-page fault.
  - TLB access: 10 ns.
  - Data access: 100 ns.
  - Total: $10 + 100 = 110$ ns.
- **Total Time**: $110 + 10,000,000 + 110 = 10,000,220$ ns.

  Marking Scheme:
  - Correct initial TLB miss (110 ns) and page fault service time (10ms): 2 marks.
  - Correct retry TLB hit (10 ns + 100 ns = 110 ns): 1 mark.
  - Incorrect or no attempt: 0 marks.